DevOps · Quick concepts

Interview revision sheet. Read it the evening before, then test yourself in Interview practice.

Core ideas

- DevOps = culture plus automation to deliver faster and safer.
- CI: merge often and build and test every change. Continuous Delivery: always releasable. Continuous Deployment: auto-release to production.
- DORA metrics: deployment frequency, lead time, change failure rate, time to restore.

Git

- clone, add, commit, push, pull (= fetch + merge).
- merge keeps history; rebase makes it linear (do not rebase shared branches).
- revert makes a new undo commit (safe); reset --hard discards changes (dangerous).
- stash saves work in progress. Trunk-based development vs GitFlow.

Docker

- Image = template; container = running instance; Dockerfile builds images.
- docker build -t app ., docker run -d, docker ps, docker logs, docker exec -it.
- Multi-stage builds give smaller images. Copy dependency files first for better caching.
- CMD = default arguments; ENTRYPOINT = main executable. Run as non-root. Avoid the latest tag.
- Volumes keep data. Docker Compose runs multi-container apps.

CI/CD tools

- Jenkins (Jenkinsfile), GitHub Actions (.github/workflows YAML), Azure Pipelines (YAML).
- Stages: build, unit test, scan (SonarQube, dependency, image), package, deploy, verify.
- Build once, deploy many; store artifacts in Nexus or Artifactory.
- Secrets in a vault or pipeline secrets, never in Git.

Infrastructure as Code

- Terraform: init, plan, apply. State file maps code to real resources; use remote state with locking.
- Modules for reuse. Drift = reality differs from code.
- Ansible: agentless, YAML playbooks, idempotent tasks.

Release strategies

- Rolling: replace gradually. Blue-green: switch traffic between two environments.
- Canary: small percentage first. Feature flags: toggle without redeploying.
- Always have a fast rollback.

GitOps and SRE

- GitOps: Git is the source of truth; Argo CD or Flux syncs the cluster.
- Shift-left and DevSecOps: test and scan early.
- Error budgets balance speed and reliability.